

Companion LXVI: Ensemble Variance of Persistence Diagrams in the Relaxation-Driven Cyclic Cosmology

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Abstract

The ensemble variance of persistence diagrams quantifies the fluctuations of topological structure across realizations of a cosmological field. In weak-lensing analyses, this variance measures the stability of connected components, loops, and void-like structures under cosmic variance, noise, and nonlinear evolution. In the standard cosmological model, the variance reflects the nonlinear matter distribution and the projection of large-scale structure. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the fluctuation spectrum of topological features in a characteristic way. This Companion develops a continuous description of ensemble variance in RDCC, deriving how the relaxation scale influences topological fluctuations, persistence-lifetime scatter, and the observational signatures in weak-lensing surveys. The resulting framework provides a distinctive probe of the RDCC gravitational field through topological variability.

1 Introduction

Persistence diagrams encode the birth and death of topological features in weak-lensing convergence fields. While Fréchet means summarize the average topology of an ensemble, the ensemble variance quantifies the fluctuations around this mean. This variance is sensitive to nonlinear structure formation, cosmic variance, and the stability of topological features.

In the standard cosmological model, ensemble variance reflects the nonlinear matter distribution and the projection of large-scale structure. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the fluctuation spectrum of topological features in a scale-dependent manner, producing a distinctive signature in ensemble-variance analyses.

2 Ensemble Variance of Persistence Diagrams

Given an ensemble of persistence diagrams $\{D_i\}$ with Fréchet mean D^* , the ensemble variance is defined as

$$\text{Var}(D) = \frac{1}{N} \sum_{i=1}^N W_p(D_i, D^*)^2, \quad (1)$$

where W_p is the p -Wasserstein distance.

This variance measures the spread of topological features across the ensemble.

3 RDCC Modification of Persistence Diagrams

In RDCC, the convergence evolves as

$$\kappa_{\text{RDCC}}(\ell) = \kappa(\ell) \left[1 - \chi_{\kappa} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (2)$$

This modifies the birth and death thresholds of topological features:

$$b_{\text{RDCC}} = b [1 - \chi_b (\ell/\ell_{\text{rel}})^{\alpha}], \quad (3)$$

$$d_{\text{RDCC}} = d [1 - \chi_d (\ell/\ell_{\text{rel}})^{\alpha}]. \quad (4)$$

The ensemble of RDCC diagrams becomes

$$\{D_{i,\text{RDCC}}\} = \{(b_{i,\text{RDCC}}, d_{i,\text{RDCC}})\}. \quad (5)$$

4 Ensemble Variance in RDCC

The RDCC ensemble variance is

$$\text{Var}_{\text{RDCC}}(D) = \frac{1}{N} \sum_{i=1}^N W_p(D_{i,\text{RDCC}}, D_{\text{RDCC}}^*)^2. \quad (6)$$

To first order in the relaxation parameter,

$$\text{Var}_{\text{RDCC}}(D) = \text{Var}(D) \left[1 - \chi_{\text{Var}} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (7)$$

This modification reflects the relaxation-driven change in topological fluctuations.

5 Topological Fluctuation Signatures of RDCC

The relaxation mechanism enhances the coherence of large-scale modes. Ensemble variance therefore exhibits:

- reduced variance at large scales (enhanced coherence),
- enhanced variance at small scales (suppressed small-scale potential),
- a characteristic turnover near ℓ_{rel} ,
- modified scaling with smoothing scale and redshift,
- altered clustering of diagrams around the Fréchet mean.

These signatures provide a direct probe of the RDCC gravitational field through topological fluctuations.

6 Conclusion

The ensemble variance of persistence diagrams in the Relaxation-Driven Cyclic Cosmology reflects the scale-dependent evolution of the gravitational potential and the nonlinear matter distribution. The relaxation mechanism modifies the fluctuation spectrum of topological features in a scale-dependent manner, producing distinctive signatures in weak-lensing surveys. These effects provide a direct observational window into the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of topological variability, nonlinear structure, and the interplay between light and gravity in RDCC.

References

- [Turner *et al.*(2014)Turner, Mileyko, Mukherjee, and Harer] K. Turner, Y. Mileyko, S. Mukherjee, and J. Harer, *Discrete and Computational Geometry* **52**, 44 (2014).
- [Mileyko *et al.*(2011)Mileyko, Mukherjee, and Harer] Y. Mileyko, S. Mukherjee, and J. Harer, *Inverse Problems* **27**, 124007 (2011).
- [Cohen-Steiner *et al.*(2007)Cohen-Steiner, Edelsbrunner, and Harer] D. Cohen-Steiner, H. Edelsbrunner, and J. Harer, *Discrete and Computational Geometry* **37**, 103 (2007).
- [Kerber *et al.*(2017)Kerber, Morozov, and Nigmetov] M. Kerber, D. Morozov, and A. Nigmetov, *Journal of Experimental Algorithmics* **22**, 1 (2017).
- [Pranav *et al.*(2017)Pranav *et al.*] P. Pranav *et al.*, *Monthly Notices of the Royal Astronomical Society* **465**, 4281 (2017).
- [Xu *et al.*(2019)Xu, Cisewski-Kehe, Green, and Nagai] X. Xu, J. Cisewski-Kehe, S. Green, and D. Nagai, *Astronomy and Computing* **27**, 34 (2019).
- [Hikage *et al.*(2008)Hikage *et al.*] C. Hikage *et al.*, *Monthly Notices of the Royal Astronomical Society* **389**, 1439 (2008).
- [Petri *et al.*(2013)Petri, Haiman, and Hui] A. Petri, Z. Haiman, and L. Hui, *Physical Review D* **88**, 123002 (2013).